

# U-AUB-02: bidefect count route, current formal frontier

Kieran McShane

2026-06-07

## U-AUB-02: bidefect count route, current formal frontier

**Status.** U-AUB-02 is still open. The current Lean declaration is a checked conditional reduction: it proves that the listed Biane, frontier, and Chapuy inputs imply `AubrunBiDefectCountBound m`; it does not prove those inputs.

### Target

The formal target behind U-AUB-02 is the corrected Aubrun count input used by the expectation pipeline. In mathematical terms, the route now tries to prove a bidefect count theorem of the following form.

For each Wick length  $m$ , every admissible pair of defects  $(a, b)$  should satisfy

$$\#\{\pi : \delta_+(\pi) = a, \delta_-(\pi) = b\} \leq E_m(a, b),$$

where  $E_m(a, b)$  is the explicit Aubrun envelope used later in the scalar sum. Once this count is known, the already checked scalar absorption turns the double defect sum into shifted bases rather than a false common-base maximum.

The remaining theorem-strength work is exactly the following four groups:

1. the large-gap left Biane adjacent-link theorem;
2. the large-gap right Biane adjacent-link theorem;
3. the side-specific residual branch collision theorems for non-fixed predecessor skips, including the nonwrapping smaller internal-subinterval child branch;
4. the Chapuy/trisection recurrence.

The active Lean endpoint is the checked reduction from those four groups to the bidefect count bound:

`AppendixB.AubrunBiDefectCountBound_of_bianeAdjacentLinkHalves_left_gap_three_start_impossibl`

This name should not be read as a solved ticket. Its theorem statement still contains the theorem-strength hypotheses listed immediately below.

The previous reduction layer was:

`AppendixB.AubrunBiDefectCountBound_of_bianeAdjacentLinkHalves_left_gap_three_start_impossible`

Delta: the previous layers still asked the collision theorem to work from the minimal-frontier packet. The active layer uses `permCyclicPredecessorSkip_direct_commonCycle` internally, supplies branch-normalized return intervals internally, descends internal-right repeats to actual shorter subintervals, splits the three origin-tag side cases, and exposes the nonwrapping internal-subinterval branch only in the off-cycle open-internal form. The newest checked support then sends the off-cycle witness through a smaller side-aware residual child and splits that child into primitive and internal-subinterval branches, then absorbs the strict primitive child alternatives into the already theorem-facing parent primitive residual leaf.

Recent left-slice improvement: the residual branch `not pi(a,a+2)` and `S(a,a+2)` is contradictory. The repeat `S(a,a+2)` copies the one-step contour move from `a+2` to `c` into a length-two path from `S(a)` to `S(c)`. Tree uniqueness compares this copied path with the direct length-two path from `pi(a,c)` and forces `T(a+2)=T(a)`, i.e. `pi(a,a+2)`, contradicting the branch assumption.

That change is U-AUB-316 in the ledger. It removes the last left-start gap-three parameter; the start slice `c-a=3`, `b=a+1` is no longer present on the endpoint surface. The later U-AUB-317 closes the right special-slice branch `S(b,c)`: if `S(c,d)` failed, copying the contour step `c -> d` across `S(b)=S(c)` would give a simple path from `S(b)` to `S(d)`, and tree uniqueness against the direct path from `pi(b,d)` would force `pi(b,c)`, contradicting `pi(a,c)` and the separating non-link `not pi(a,b)`.

The active endpoint reduces `AubrunBiDefectCountBound m` to the following theorem-strength inputs:

1. the remaining large-gap left Biane adjacent link under  $2 < c-a$  and  $c-a \neq 3$ ;
2. the remaining large-gap right Biane adjacent link outside the checked  $d-b=3$ ,  $c=b+2$  slice;
3. the side-specific residual branch collision theorems for non-fixed predecessor skips, together with the strict nonwrapping smaller child internal-subinterval collision theorem carrying the parent/child side data;
4. the Chapuy/trisection recurrence.

Current support discipline: these live leaf groups are also rendered as a Graphviz frontier at `build/u_aub_02_frontier.svg`, tracked in the leanblueprint declaration inventory, and checked by `tools/build_formalization_artifacts.sh`. Grok review is used only as scientific criticism of the exposition; it is not a second formalization agent. The finite permutation scripts remain conjecture finders only.

Previous delta: U-AUB-349 consumes the off-cycle witness through a smaller side-aware residual interval. The smaller contour-repeat interval now splits: either it already gives the common  $\pi/S\pi$  pair, or it leaves a strictly smaller **leftTightStrictReturnIntervalResidual** with the inherited no- $x$ -cycle and block-off- $x$  invariants. The current endpoint therefore no longer asks for the whole nonwrapping off-cycle open-internal subinterval theorem.

Most recent delta: U-AUB-358 absorbs the strict child primitive branches into the parent primitive residual leaf. The theorem face therefore no longer asks separately for the child internal **S**, start **S**, end **S**, fixed predecessor, or adjacent **T** branches. Those are all primitive residuals on a strict child interval, hence primitive residuals on the parent interval. The remaining smaller nonwrapping child leaf is the genuine internal-subinterval branch.

Previous delta: U-AUB-355 wires the strict split of the smaller contour-repeat packet into the sharp endpoint. U-AUB-356 then replaces raw internal **T** by adjacent **T** or genuine shorter internal subinterval, and U-AUB-358 absorbs the primitive alternatives as above.

Previous delta: U-AUB-350 splits that smaller child residual by the subinterval-preserving descent. The theorem face now asks separately for the side-aware primitive child branch and the side-aware internal-subinterval child branch; the raw child residual predicate is no longer the exposed input.

Previous delta: U-AUB-348 turns the confined off-cycle internal witness into a strictly smaller charged contour interval. The witness endpoints  $a + r, a + s$  form a same- $\pi$  interval strictly inside  $(a, c)$ , the interval has its own **leftTightContourIntervalRepeat** packet, and the whole witness block is still off the skipped  $x$ -cycle. This is support for the open collision leaf, not a closure of that leaf.

Previous delta: U-AUB-347 proves strict noncrossing confinement for the off-cycle nonwrapping internal block. If  $a, c$  are the first-return endpoints in the skipped  $x$ -cycle and the internal witness starts at  $a + r$  off that cycle, the whole  $\pi$ -block of  $a + r$  lies strictly inside  $(a, c)$ .

Previous delta: U-AUB-329 makes the branch-normalized return interval theorem-facing at the current sharp endpoint. The collision theorem no longer has to consume an arbitrary minimal-frontier packet; it can use the nonwrapping first-return, wrap-below last-return, or wrap-above first-return interval tied to the original skip. This is a sharper checked conditional endpoint, not a proof of the collision theorem.

Previous delta: U-AUB-328 completes the branch-normalized return-interval package for nested predecessor skips. The wrap-below branch now chooses the last same- $\pi$  return below  $x$ ; because a nonfixed wrapping skip has  $x.val < (\pi x).val$ , the interval ending at  $(\pi x).val$  has a genuine gap and carries the repeated-contour obstruction. The unified theorem packages nonwrapping first return, wrap-below last re-

turn, and wrap-above first return. This still does not prove the nested collision theorem. A new pure-math article source was also created at `article/u_aub_02_pure_math_article.tex`; it compiles cleanly and has one Grok referee pass recorded in `build/grok_u_aub_02_article_referee.md`.

Previous delta: U-AUB-327 extracts first-return contour intervals in two branches of the nested predecessor-skip leaf. In the nonwrapping branch, the first same- $\pi$  return after  $(\pi\ x).val$  is separated by a genuine gap unless the already-closed direct-hit case occurs. The same first-return package is now available for the wrap-above branch. Both packages include the left-tight repeated-contour obstruction. They do not prove the collision theorem.

Previous delta: U-AUB-326 adds a tied contour extractor for the nested predecessor-skip leaf. A non-fixed skip now produces a long repeated contour interval together with an origin tag: either the interval is exactly  $[(\pi\ x).val, x.val]$ , or it is a wrapping interval ending at  $(\pi\ x).val$ , or it is a wrapping interval starting at  $x.val$ . This prevents the later collision proof from treating the charged interval as anonymous. It does not prove the collision theorem.

Previous delta: U-AUB-325 removes the direct-hit branch from the minimal-frontier leaf. If `finRotate`  $(\pi\ x)$  is already in the same  $\pi$ -cycle as  $x$ , the checked theorem `permCyclicPredecessorSkip_direct_commonCycle` supplies the nontrivial common  $\pi/\text{finRotate}*\pi$  pair. The live frontier theorem therefore only needs to handle the nested case where this same-cycle relation fails.

Most recent delta, right special slice: U-AUB-320 closes the last checked right branch,  $S(b+1, d)$ . If  $S(b+1, d)$  holds while the target  $S(c, d)$  fails, uniqueness of simple incidence paths from  $S(c)$  to  $S(d)$  forces  $\pi(b+1, c)$ . Combined with  $\pi(a, c)$ , this gives  $\pi(a, b+1)$ . The shifted left adjacent-link input on  $(a, b, b+1, d)$  gives  $S(a, b)$ , and then  $S(a, b)$ ,  $S(b+1, d)$ ,  $\pi(a, b+1)$ , and  $\pi(b, d)$  form the forbidden incidence square. The active endpoint no longer exposes `hRightS_b1_d`, `hRightFixed_b1`, `hRightS_b_c`, or `hRightT_b1_c`. This is a checked closure of the right gap-three special slice; U-AUB-02 remains open because the non-local leaves below are still theorem-facing.

Recent delta, left end slice: U-AUB-321 closes the fixed  $a+1$  branch. In the slice  $c=a+3$ ,  $b=a+2$ , fixedness of  $a+1$  gives the adjacent repeat  $S(a+1, b)$ . If the target  $S(a, b)$  failed, the contour step  $a \rightarrow a+1$ , this repeat, and the contour step  $b \rightarrow c$  would form a simple length-four incidence path from  $S(a)$  to  $S(c)$ . Tree uniqueness compares that path with the direct length-two path from  $\pi(a, c)$ , contradiction. The active endpoint no longer exposes `hLeftEndFixed_a1`.

Most recent delta, left end slice: U-AUB-322 closes the remaining  $S(a+1, c)$  branch. The branch copies the contour step  $a+1 \rightarrow b$  into a simple path from  $S(c)$  to  $S(b)$ . The reversed contour step  $b \rightarrow c$  gives a second simple path with the same endpoints. Tree uniqueness identifies their middle right vertices, hence forces  $\pi(a+1, b)$ . The existing  $\pi(a+1, b)$  absorber then closes the target. The active endpoint no longer exposes any finite gap-three branch.

Tooling delta: U-AUB-323 makes the large-gap sanity profile reproducible. The branch sanity script now checks the live large-gap crossing hypotheses in small left-tight models; through `n=9`, it enumerates 6,916 left-tight permutations and finds no crossing hypotheses. This is conjecture-finding evidence only, not a substitute for the large-gap Biane theorem.

This endpoint is a checked reduction, not a no-input closure of U-AUB-02. Each item in the list above is still theorem-facing. Thus a `leanok` marker for this endpoint means that the reduction from these inputs is verified; it does not mean that the Aubrun count theorem has been proved unconditionally. In particular, words such as `Biane`, `frontier`, or `chapuy` inside the Lean name are not completion markers. They name the hypotheses being consumed.

## Mathematical route

This section states the intended counting theorem, not a currently closed checked theorem.

The corrected route is one-sided first. The intended combinatorial theorem interprets the plus-defect  $\delta_+$  through the unicellular-map genus:

$$\delta_+(\pi) = 2g(\pi).$$

The expected counting strategy is then:

$$\#\{\delta_+ = 0\} \leq 4^{m+1},$$

and for genus  $g$ ,

$$N_{m,g+1} \leq (2(m+1))^3 N_{m,g}.$$

Iterating gives

$$N_{m,g} \leq 4^{m+1} (2(m+1))^{3g}.$$

This is the combinatorial content that should feed the bidefect envelope. This paragraph is a proof target, not a closed checked theorem. The current verified content is the reduction from concrete Biane/Chapuy inputs to the bidefect envelope; the unconditional genus-zero and Chapuy counting theorems remain open.

## Current Checked Content

The proof graph contains a large amount of checked surrounding structure.

The high plus-defect tail, finite small-window reductions, bidefect scalar absorption, `NCPart` cardinality, gap-two Biane adjacent links, and several minimal-contour/frontier reductions are already verified. The latest checked endpoint sharpens the route as follows:

large-gap Biane half-links  
+ checked gap-three frontier reductions

+ side-specific residual branch collisions  
+ Chapuy recurrence  
=> AubrunBiDefectCountBound m.

This display is a conditional implication, not a completed theorem. The named declaration still carries the large-gap Biane inputs, the side-specific residual return-interval collision theorems, and the Chapuy recurrence as hypotheses.

As a conditional adapter, the endpoint narrows the previous large-gap route in two ways. First, it removes the direct no-skip hypothesis from the theorem face. Instead of asking directly that a noncrossing left-tight cycle partition has no non-fixed predecessor skip, the current endpoint asks for the sharper collision statement:

If a non-fixed predecessor skip exists, extract the side-tied long same- $\pi$  return interval. Split the origin side into nonwrapping first return, wrap-below last return, and wrap-above first return; then split each side into primitive and internal-subinterval branches. In the nonwrapping internal-subinterval branch the endpoint-touching case and the case where the internal witness remains in the skipped  $x$ -cycle are checked impossible, so the theorem-facing nonwrapping child leaf is now split into the side-aware primitive and internal-subinterval child branches produced from the off-cycle witness; the child interval lies strictly inside the first-return interval, has no skipped  $x$ -cycle points inside it, and its  $\pi$ -block stays inside the parent interval and off  $x$ . The remaining branch theorems must force two distinct points  $y, z$  such that

$$y \sim_{\pi} z \quad \text{and} \quad y \sim_{\gamma\pi} z.$$

This contradicts left-tight transversality.

Second, it no longer asks the broad left large-gap Biane hypothesis to handle any  $c-a=3$  slice. Since  $a < b < c$  and  $c-a=3$  force either  $b=a+1$  or  $b=a+2$ , both left length-three slices are now routed through finite contour reductions.

For  $c-a=3$ ,  $b=a+1$ , the left start-slice branches are now all discharged at the active endpoint. The four former branches were

$$S(b, c), \quad \pi(a+2) = a+2, \quad \pi(b, a+2), \quad (\neg(a \sim_{\pi} a+2) \wedge S(a, a+2)).$$

The left-start  $S(b, c)$  among them is supplied by the right Biane frontier. The fixed-point branch and the  $\pi(b, a+2)$  branch both force that already-closed left-start  $S(b, c)$  branch. The last branch is closed directly by the copied length-two incidence path described above.

For  $c-a=3$ ,  $b=a+2$ , the remaining left finite branches are

$$S(a+1, c).$$

The earlier start-slice proof used several normalizations before the final closure. If the raw branch  $S(a, a+2)$  also has the hidden link  $a \sim_{\pi} a+2$ ,

the checked start-gap-two right-repeat theorem gives the target  $S(a, b)$ . The adjacent branch  $S(a + 2, c)$  is exactly the adjacent repeat at  $a + 2$ , hence becomes the fixed-point branch  $\pi(a + 2) = a + 2$ . The target  $S(a, b)$  itself is impossible:  $b = a + 1$ , while  $a$  already lies in a nontrivial same- $\pi$  interval with  $c$  and  $c - a = 3$ . An adjacent  $(\gamma\pi)$ -repeat  $S(a, a + 1)$  would force  $a$  to be fixed by  $\pi$ , contradicting  $a \sim_\pi c$  with  $a < c$ . The checked theorem `leftTight_bianeAdjacentLink_left_of_gap_three_start_adjacent_branch_to_target_iff_false` therefore records that branch-to-target implications in this slice are the same as proving the corresponding branch impossible. The branch  $\pi(a, a + 1)$  is now closed: together with  $\pi(a, c)$  it gives  $\pi(a + 1, c)$ , so the shifted gap-two crossing  $(a + 1, b, c, d)$  gives  $S(a + 1, b)$ , which feeds the already exposed branch. The branch  $\pi(a + 1, b)$  is also closed: viewed on the shifted crossing  $(a, a + 1, c, d)$ , it is the already exposed start-adjacent  $\pi(b, a + 2)$  branch. That branch gives  $S(a, a + 1)$ , which is impossible for the nontrivial same- $\pi$  interval  $a \sim_\pi c$ . The remaining adjacent-repeat branch  $S(a + 1, b)$  has also been normalized: because  $b = a + 2$ , it is exactly the adjacent left repeat at  $a + 1$ , and left-tightness turns it into the fixed-point branch  $\pi(a + 1) = a + 1$ . That fixed-point branch is now closed: fixedness gives  $S(a + 1, b)$ , and a denial of  $S(a, b)$  produces a simple length-four path from  $S(a)$  to  $S(c)$ . Tree uniqueness compares it with the direct length-two path from  $a \sim_\pi c$ , contradiction.

The right branch  $S(b, c)$  is now closed: assuming  $S(b, c)$  and denying the target  $S(c, d)$  gives a copied contour path from  $S(b)$  to  $S(d)$ , and tree uniqueness against the direct  $\pi(b, d)$  path forces the forbidden link  $\pi(b, c)$ . This right special branch is no longer a theorem-facing input to the active endpoint.

The branch  $\pi(b + 1, c)$  is also no longer independent. It combines with the crossing link  $\pi(a, c)$  to give  $\pi(a, b + 1)$ . Applying the large-gap right input to the shifted crossing  $(a, b, b + 1, d)$  gives  $S(b + 1, d)$ , because the shifted crossing is not the special  $c = b + 2$  slice.

The fixed branch  $\pi(b + 1) = b + 1$  is now also internal. Fixedness gives the adjacent repeat  $S(b + 1, c)$ . If  $S(b + 1, d)$  failed, the contour step  $b \rightarrow b + 1$  followed by the copied contour step  $c \rightarrow d$  would be a simple length-four incidence path from  $S(b)$  to  $S(d)$ . Tree uniqueness compares that path with the direct length-two path from  $\pi(b, d)$ , contradiction. So the fixed branch feeds the same  $S(b + 1, d)$  implication.

The last right branch  $S(b + 1, d)$  is now closed. If the target  $S(c, d)$  were false, path uniqueness would force  $\pi(b + 1, c)$ . Then  $\pi(a, c)$  gives  $\pi(a, b + 1)$ , the shifted left link gives  $S(a, b)$ , and the four incidences  $\pi(a, b + 1)$ ,  $\pi(b, d)$ ,  $S(a, b)$ ,  $S(b + 1, d)$  form a forbidden square.

There are no remaining right finite branches in the checked  $d - b = 3$ ,  $c = b + 2$  slice. The right branch  $\pi(b, c)$  is now closed: together with the crossing link  $\pi(a, c)$ , it would imply  $\pi(a, b)$ , contradicting the separating non-link. The adjacent-repeat branch  $S(b + 1, c)$  has also been normalized: because  $c = b + 2$ , it is exactly the adjacent left repeat at  $b + 1$ , and left-tightness turns it into the fixed-point

branch  $\pi(b+1) = b+1$ .

These branch lists are endpoint-local snapshots. The same public endpoint still separately requires the large-gap left Biane input, the large-gap right Biane input, the side-specific residual collision theorems, and the Chapuy recurrence.

## What remains open

U-AUB-02 is not done. The remaining theorem-strength leaves are:

1. prove the remaining large-gap left Biane adjacent-link cases;
2. prove the remaining large-gap right Biane adjacent-link cases;
3. prove the remaining side-specific residual collision theorems, now with branch-normalized return data available in all three nested branches and the nonwrapping child residual carrying the inherited no- $x$ -cycle and block-off- $x$  data; only the genuine smaller child internal-subinterval branch remains theorem-facing after child primitive absorption;
4. prove the Chapuy/trisection recurrence.

These branches are not mere notation. They are finite Biane contour cases that likely need incidence-tree uniqueness or a descent lemma, not a naive cycle-transitivity shortcut. The finite branch scripts are conjecture finders only; their clean output is not counted as proof.

## Checked reduction evidence

The new endpoint is in:

`PptFactorization/AppendixBAubrunGraduate.lean`

The active checked declaration is:

`PptFactorization.AppendixB.AubrunBiDefectCountBound_of_bianeAdjacentLinkHalves_left_gap_thre`

The previous reduction layer is:

`PptFactorization.AppendixB.AubrunBiDefectCountBound_of_bianeAdjacentLinkHalves_left_gap_thre`

Verification performed:

```
python3 scripts/lean_quick_check.py PptFactorization.AppendixBAubrunGraduate --timeout 300
lake build PptFactorization.AppendixBAubrunGraduate
lake build PptFactorization.AristotleTargets.UpperFromLowerColumnConcreteChoices
python3 scripts/aubrun_branch_sanity.py --max-n 9 --all
Pandoc rebuild of exposition/u_aub_02_formalization_note.pdf
Blueprint print-PDF rebuild and declaration inventory check
Grok adversarial review
#print axioms ...left_gap_three_start_impossible_frontiers_nestedReturnContourIntervalCommon
#print axioms ...left_of_gap_three_end_adjacent_fixed_a1
```

The axiom audit reports only:



[propext, Classical.choice, Quot.sound]

This audit is evidence that the current reduction layer has no project-specific axioms. It is not evidence for the open leaves themselves. In particular, the large-gap Biane links, the nested non-direct return-interval collision theorem, and the Chapuy recurrence still need their own checked declarations and axiom audits.

## Tooling status

Automatic prose narration is not used as proof evidence. The current note is hand-curated from the checked Lean declarations, and any future machine-written prose must be audited against the exact hypothesis list of the active endpoint.

Pandoc is installed and converts this note to both LaTeX and PDF using the repository TeX setup. This PDF has been refreshed from the current Markdown. Leanblueprint was initialized for the project, with a declaration inventory at `blueprint/lean_decls`; blueprint refreshes are treated as a separate verification step and are not proof evidence by themselves. Full doc-gen integration was deliberately deferred when it attempted to download a 518 MB Lean release-candidate toolchain, which would slow the current proof workflow.

A finite branch sanity script is now available at `scripts/aubrun_branch_sanity.py`. It enumerates small left-tight permutations and checks active branch implications for counterexamples before they are attempted in Lean. This is only a guardrail: the current script reports no active finite gap-three branches, then records the large-gap profile through `n=9`.

Grok was used for advisory proof-architecture and exposition review. Its commentary remains advisory; Lean declarations, builds, axiom audits, Pandoc artifacts, and blueprint checks are the evidence.